

Exponent (Indices) Laws

Objectives

- Evaluating numerical expressions with a positive or negative fractional exponent
- Writing numerical expressions with fractional exponents as radicals
- Using the rules of **indices** to simplify expressions that contain radicals and/or fractional exponents

Inquiry questions

- F** • How do you evaluate a number with a fractional exponent?
- How do you use the rules of **indices** to simplify expressions with fractional exponents?
- C** • How are the rules of **indices** and the rules of radicals related?
- How do the rules of **indices** help simplify radical expressions?

ATL Critical-thinking

Analyse complex concepts and projects into their constituent parts and synthesize them to create new understanding

You should already know how to:

• use the multiplicative rule for exponents	1 Simplify: a $(x^3)^2$ b $(2^4)^2$ c $(2x^2)^3$
• use the rules of radicals to simplify expressions	2 Simplify completely: a $\sqrt{45}$ b $\frac{\sqrt{128}}{\sqrt{8}}$ c $\sqrt{60} \times \sqrt{3}$
• evaluate expressions with negative exponents	3 Evaluate, leaving your answer as a fraction where necessary: a 4^{-1} b 6^{-2} c 2^{-3} d $\left(\frac{1}{2}\right)^{-1}$ e $\left(\frac{2}{3}\right)^{-1}$ f $\left(\frac{2}{5}\right)^{-2}$

Solutions:

You should already know how to:

1 a x^6

b 2^8

c $8x^6$

2 a $3\sqrt{5}$

b 4

c $6\sqrt{5}$

3 a $\frac{1}{4}$

b $\frac{1}{36}$

c $\frac{1}{8}$

d 2

e $\frac{3}{2}$

f $\frac{25}{4}$

Review: Laws of Exponents (Indices)

If a and b are real numbers and m and n are positive integers, we have that

Law	Rule	Example
1. Multiplication [same base]	$a^m \times a^n = a^{m+n}$	$3^4 \times 3^6 = 3^{4+6} = 3^{10}$
2. Division [same base]	$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$	$7^9 \div 7^5 = 7^{9-5} = 7^4$
3. Power of a power [same base]	$(a^m)^n = a^{m \times n}$	$(2^3)^5 = 2^{3 \times 5} = 2^{15}$
4. Power of a power [same power]	$a^m \times b^m = (ab)^m$	$3^4 \times 7^4 = (3 \times 7)^4 = 21^4$
5. Division [same power]	$a^m \div b^m = \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$	$5^3 \div 7^3 = \left(\frac{5}{7}\right)^3$
6. Negative one to a power	$(-1)^n = \begin{cases} -1 & \text{if } n \text{ is odd} \\ 1 & \text{if } n \text{ is even} \end{cases}$	$(-1)^3 = (-1)^5 = \dots = -1$ $(-1)^2 = (-1)^4 = \dots = 1$

Review Questions: Simplify these expressions

a) $\frac{15y^{-4} \times 24y^{10}}{6y^7} =$

b) $(s^6)^{-2} \times 4s^{-6} =$

c) $(5a^3b^5)^2 \times 4a^2b^2 =$

d) $(s^{-3}t)^{-2} \div 3s^{-2}t^5 =$

e) $\frac{15r^{-2} \times 20r^{12}}{12r^9} =$

f) $\frac{(6a^2b^7)^3 \times 5a^4b^3}{3a^5b} =$



g) $\frac{21y^{-4} \times 16y^{-11}}{6y^8} =$

h) $\frac{(s^4)^{-3} \times (7s^{-2}t)}{(2st)^2} =$

i) $\frac{18z^{-4} \times 16z^{11}}{6z^8} =$

j) $(5x^7y^3z^{-1})^2 \cdot (2xy^{-5})^3 \cdot (2y^{-3}z^2)^3$

k) Simplify fully $\frac{\left(2x^{\frac{1}{2}}\right)^3}{4x^2}$

Solutions:

$$a) \frac{15y^{-4} \times 24y^{10}}{6y^7} = 60y^{-4+10-7} = 60y^{-1} = \boxed{\frac{60}{y}}$$

$$b) (s^6)^{-2} \times 4s^{-6} = s^{-12} \cdot 4s^{-6} = 4s^{-18} = \boxed{\frac{4}{s^{18}}}$$

$$c) (5a^3b^5)^2 \times 4a^2b^2 = 25a^6b^{10} \cdot 4a^2b^2 = \boxed{100a^8b^{12}}$$

$$d) (s^{-3}t)^{-2} \div 3s^{-2}t^5 = \frac{s^6t^{-2}}{3s^{-2}t^5} = \boxed{\frac{s^8}{3t^7}}$$

$$e) \frac{15r^{-2} \times 20r^{12}}{4r^9} = \boxed{25r}$$

$$f) \frac{(6a^2b^7)^3 \times 5a^4b^3}{3a^5b} = \boxed{360a^5b^{23}}$$

$$g) \frac{21y^{-4} \times 16y^{-11}}{6y^8} = \boxed{\frac{56}{y^{23}}}$$

$$h) \frac{(s^4)^{-3} \times (7s^{-2}t)}{(2st)^2} = \boxed{\frac{7}{4s^{16}t}}$$

$$i) \frac{18z^{-4} \times 16z^{11}}{6z^8} = \boxed{\frac{48}{z}}$$

$$j) (5x^7y^3z^{-1})^2 \cdot (2xy^{-5})^3 \cdot (2y^{-3}z^2)^3 = \boxed{\frac{1600x^{17}z^4}{y^{18}}}$$

$$k) \text{ Simplify fully } \frac{(2x^{\frac{1}{2}})^2}{4x^2} = 2x^{-\frac{1}{2}} = \boxed{\frac{2}{x^{\frac{1}{2}}}}$$

F Fractional exponents

- How do you evaluate a number with a fractional exponent?
- How do you use the rules of indices to simplify expressions with fractional exponents?

For any positive real number x , and integer n , where $n \neq 0$, $x^{\frac{1}{n}} = \sqrt[n]{x}$.

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Example 1

Evaluate:

a $\left(\frac{9}{25}\right)^{\frac{1}{2}}$

b $\left(\frac{1}{8}\right)^{-\frac{1}{3}}$

c $\left(\frac{7}{81}\right)^{-\frac{1}{2}}$

a $\left(\frac{9}{25}\right)^{\frac{1}{2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$

b $\left(\frac{1}{8}\right)^{-\frac{1}{3}} = 8^{\frac{1}{3}} = \sqrt[3]{8} = 2$

c $\left(\frac{7}{81}\right)^{-\frac{1}{2}} = \left(\frac{81}{7}\right)^{\frac{1}{2}} = \frac{9}{\sqrt{7}} = \frac{9\sqrt{7}}{7}$

Using the multiplicative rule for exponents,

$$(x^a)^b = x^{ab}: \left(\frac{1}{8}\right)^{-\frac{1}{3}} = \left(\left(\frac{1}{8}\right)^{-1}\right)^{\frac{1}{3}} = 8^{\frac{1}{3}}$$

Here, the denominator has been rationalized.



Practice 1

1 Work out the value of:

a $\left(\frac{1}{4}\right)^{\frac{1}{2}}$

b $\left(\frac{1}{16}\right)^{-\frac{1}{4}}$

c $\left(\frac{1}{8}\right)^{\frac{1}{3}}$

d $\left(\frac{1}{25}\right)^{-\frac{1}{2}}$

e $\left(\frac{4}{9}\right)^{\frac{1}{2}}$

f $27^{-\frac{1}{3}}$

g $\left(\frac{4}{25}\right)^{-\frac{1}{2}}$

h $\left(\frac{27}{1000}\right)^{-\frac{1}{3}}$

2 Simplify and leave your final answer as a power of x :

a $(4x)^{\frac{1}{2}}$

b $(27x^3)^{\frac{1}{3}}$

c $(8x^6)^{-\frac{1}{3}}$

d $\left(\frac{x^4}{100}\right)^{-\frac{1}{2}}$

e $x^{-\frac{1}{2}} \times x^{\frac{1}{2}}$

f $x^{\frac{1}{3}} \div x^{\frac{1}{2}}$

g $x^{\frac{1}{4}} \times x^{-\frac{1}{2}}$

h $(9x^2)^{-\frac{1}{2}} \times 9x^2$

Assume that $x \neq 0$
for all expressions.

Problem solving

3 Write each of these using exponents instead of the radicals.

a $\frac{1}{\sqrt{x}}$

b $\sqrt[3]{x}$

c $\frac{1}{\sqrt[4]{x}}$

4 Solve:

a $\left(\frac{1}{x}\right)^{-\frac{1}{3}} = 2$

b $\left(\frac{y}{9}\right)^{-\frac{1}{2}} = 0.3$

Solutions:

Practice 1

1 a $\frac{1}{2}$

b 2

c $\frac{1}{2}$

d 5

e $\frac{2}{3}$

f $\frac{1}{3}$

g $\frac{5}{2}$

h $\frac{10}{3}$

2 a $2x^{\frac{1}{2}}$

b $3x$

c $\frac{1}{2x^2}$

d $\frac{10}{x^2}$

e 1

f $x^{-\frac{1}{6}}$

g $x^{-\frac{1}{4}}$

h $3x$

3 a $x^{-\frac{1}{2}}$

b $x^{\frac{1}{3}}$

c $x^{-\frac{1}{4}}$

4 a $x = 8$

b $y = 100$

Using the multiplicative rule for exponents $(x^a)^b = x^{ab}$:

$$(x^2)^3 = x^{\frac{1}{2}}$$

$$(x^4)^3 = x^{\frac{4}{3}}$$

Reflect and discuss 1

What do expressions with fractional exponents like $x^{\frac{2}{3}}$ or $x^{\frac{4}{3}}$ mean?

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Exploration 1

1 Copy and complete this table:

Using integer and unit fractional exponents	Using a radical	Using fractional exponents
$\left(x^{\frac{1}{2}}\right)^3$	$(\sqrt{x})^3$	$x^{\frac{3}{2}}$
$\left(x^{\frac{1}{2}}\right)^5$		
$(x^3)^{\frac{1}{2}}$	$\sqrt{x^3}$	
	$\sqrt[3]{x^4}$	
		$x^{\frac{2}{5}}$
		$x^{\frac{2}{3}}$

A unit fraction has numerator 1.

2 Simplify $\left(x^{\frac{1}{3}}\right)^2$ and $(x^2)^{\frac{1}{3}}$. Are $\left(x^{\frac{1}{3}}\right)^2$ and $(x^2)^{\frac{1}{3}}$ equivalent?

Generalize your result for $\left(x^{\frac{1}{n}}\right)^m$ where m and n are integers and $n \neq 0$.

3 Copy and complete, using your result from step 2:

$$(\sqrt[n]{x})^m = \left(x^{\frac{1}{n}}\right)^m = x^{\frac{m}{n}} = (x^m)^{\frac{1}{n}} = \sqrt[n]{x^m}$$



Reflect and discuss 2

- Did everyone in the class get the same results for the last two rows in the table in Exploration 1?
 - Is there more than one way of writing $x^{\frac{2}{3}}$ and $x^{\frac{3}{4}}$ in the other two columns?
- 

Solution:

Exploration 1

1

Using integer and unit fractional exponents	Using a radical	Using fractional exponents
$\left(x^{\frac{1}{2}}\right)^3$	$(\sqrt{x})^3$	$x^{\frac{3}{2}}$
$\left(x^{\frac{1}{2}}\right)^5$	$(\sqrt{x})^5$	$x^{\frac{5}{2}}$
$(x^3)^{\frac{1}{2}}$	$\sqrt{x^3}$	$x^{\frac{3}{2}}$
$(x^4)^{\frac{1}{3}}$	$\sqrt[3]{x^4}$	$x^{\frac{4}{3}}$
$\left(x^{\frac{1}{5}}\right)^2$ or $(x^2)^{\frac{1}{5}}$	$(\sqrt[5]{x})^2$ or $\sqrt[5]{x^2}$	$x^{\frac{2}{5}}$
$\left(x^{\frac{1}{3}}\right)^2$ or $(x^2)^{\frac{1}{3}}$	$(\sqrt[3]{x})^2$ or $\sqrt[3]{x^2}$	$x^{\frac{2}{3}}$

2 $\left(x^{\frac{1}{3}}\right)^2 = x^{\frac{2}{3}} = (x^2)^{\frac{1}{3}}$ Their results are equal, but the process to obtain the result is different.

3 $(\sqrt[n]{x})^m = \left(x^{\frac{1}{n}}\right)^m = x^{\frac{m}{n}} = (x^m)^{\frac{1}{n}} = \sqrt[n]{x^m}$

Index Form to Surd Form conversion

For any positive real number x , and integers m and n , where $n \neq 0$,

$$x^{\frac{m}{n}} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$$

Recall how to express negative indices also.

$$a^{-m} = \frac{1}{a^m}$$

Example 2

Simplify $16^{\frac{3}{4}}$ without using a calculator.

$$16^{\frac{3}{4}} = (16^{\frac{1}{4}})^3$$

Split the exponent into a unit fraction and an integer.

$$= (2)^3$$

$$= 8$$

$$16^{\frac{1}{4}} = \sqrt[4]{16} = 2$$

Reflect and discuss 3

You could also simplify $16^{\frac{3}{4}}$ like this: $16^{\frac{3}{4}} = (16^3)^{\frac{1}{4}}$. Complete the calculation and verify that this gives the same answer as Example 2.

Which was easier – this method, or the one in Example 2? Explain why.

Write a hint to help you simplify expressions with fractional exponents.



Reflect and discuss 3

$(16^3)^{\frac{1}{4}} = \sqrt[4]{4096} = 8$ gives the same answer.

The first method is easier because it is easier to work with smaller numbers than with larger numbers. A hint to make it easier to evaluate $a^{\frac{m}{n}}$ would be to first take the n^{th} root of a before raising it to the power of m .

You can choose what method to use, based on the type of problem you have to solve, or your preferred method.

Mixed Examples for your reference:

$$\begin{aligned} \text{a} \quad & 49^{\frac{1}{2}} \\ &= \sqrt{49} \\ &= 7 \end{aligned}$$

$$\begin{aligned} \text{b} \quad & 27^{\frac{1}{3}} \\ &= \sqrt[3]{27} \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{c} \quad & 49^{-\frac{1}{2}} \\ &= \frac{1}{49^{\frac{1}{2}}} \\ &= \frac{1}{\sqrt{49}} \\ &= \frac{1}{7} \end{aligned}$$

$$\begin{aligned} \text{d} \quad & 27^{-\frac{1}{3}} \\ &= \frac{1}{27^{\frac{1}{3}}} \\ &= \frac{1}{\sqrt[3]{27}} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{a} \quad & 8^{\frac{4}{3}} \\ &= (2^3)^{\frac{4}{3}} \\ &= 2^{3 \times \frac{4}{3}} \\ &= 2^4 \\ &= 16 \end{aligned}$$

$$\{(a^m)^n = a^{mn}\}$$

$$\begin{aligned} \text{b} \quad & 27^{-\frac{2}{3}} \\ &= (3^3)^{-\frac{2}{3}} \\ &= 3^{3 \times -\frac{2}{3}} \\ &= 3^{-2} \\ &= \frac{1}{9} \end{aligned}$$

Practice 2



1 Simplify these expressions without using a calculator.

a $8^{\frac{2}{3}}$

b $27^{\frac{4}{3}}$

c $25^{\frac{3}{2}}$

d $64^{\frac{2}{3}}$

e $125^{\frac{2}{3}}$

f $343^{\frac{2}{3}}$

Problem solving

2 Write each of these using exponents instead of the radicals.

a $\sqrt[3]{x^2}$

b $(\sqrt[4]{x})^3$

c $(\sqrt{x})^3$

d $\sqrt{x^5}$

3 Simplify these. Where applicable, leave your final answer as a power of x .

a $(x^{\frac{2}{3}})^2$

b $x^{\frac{1}{3}} \times x^4$

c $x^{\frac{1}{2}} \times x^{\frac{2}{3}}$

d $(3^4)^{\frac{1}{2}}$

e $(x^{\frac{2}{5}})^{\frac{2}{3}}$

f $(32^2)^{\frac{2}{5}}$



Practice 3

1 Find the value of each expression. Leave your answer as a simplified fraction.

a $25^{-\frac{3}{2}}$

b $16^{-\frac{3}{4}}$

c $\left(\frac{9}{16}\right)^{-\frac{3}{2}}$

d $\left(\frac{1}{64}\right)^{-\frac{4}{3}}$

e $\left(\frac{36}{25}\right)^{-\frac{3}{2}}$

f $\left(\frac{81}{16}\right)^{-\frac{3}{4}}$

2 Simplify these expressions completely.

a $125^{\frac{1}{4}} \times 125^{\frac{1}{12}}$

b $16^{\frac{1}{6}} \times 16^{\frac{1}{12}}$

c $49^{\frac{3}{2}} \times 343^{\frac{1}{3}}$

d $8^{\frac{3}{4}} \div 8^{\frac{1}{12}}$

e $27^2 \div 27^{\frac{2}{3}}$

f $45^{\frac{1}{2}} \div 15^{\frac{1}{2}}$

Problem solving

3 Write down an equivalent expression that contains fractional exponents:

a 2

b 5

c $\frac{1}{2}$

d $\frac{2}{3}$

4 Simplify:

a $3^2 \times 9^{-2}$

b $4^{\frac{1}{4}} \times 2^{\frac{3}{2}}$

c $3^{\frac{1}{2}} \times 27^{\frac{1}{2}}$

d $16^{\frac{1}{3}} \div 2^{\frac{1}{3}}$

In question 4, first write both numbers as powers of the same base.

For example:
 $3^2 \times 9^{-2} = 3^2 \times (3^2)^{-2}$

Solutions:

Practice 2

1 a 4

b 81

c 125

d 16

e 25

f 49

2 a $x^{\frac{2}{3}}$

b $x^{\frac{3}{4}}$

c $x^{\frac{3}{2}}$

d $x^{\frac{5}{2}}$

3 a $x^{\frac{4}{3}}$

b $x^{\frac{13}{3}}$

c $x^{\frac{7}{6}}$

d $3^2 = 9$

e $x^{\frac{4}{15}}$

f 16

Practice 3

1 a $\frac{1}{125}$

b $\frac{1}{8}$

c $\frac{64}{27}$

d 256

e $\frac{125}{216}$

f $\frac{8}{27}$

2 a 5

b 2

c 2401

d 4

e 81

f $\sqrt{3}$

3 Student's own answers

4 a $\frac{1}{9}$

b 4

c 9

d 2

C Rules of indices and radicals

- How are the rules of indices and the rules of radicals related?
- How do the rules of indices help simplify radical expressions?

Writing a radical as an exponent can help you simplify some expressions.

Example 4

Write $32\sqrt{2}$ as a power of 2.

$$\begin{aligned} 32\sqrt{2} &= 2^5 \times 2^{\frac{1}{2}} \\ &= 2^{\frac{11}{2}} \end{aligned}$$

Write each number as a power of 2, then simplify.

$$\text{For } a, b \geq 0: \sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}.$$

$$\text{For } a \geq 0, b > 0: \sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}.$$

$$\text{For } a \geq 0, n > 0: \sqrt[n]{a^n} = a.$$



Write each of the following as powers of 2.

1. 8

2. 32

3. $\frac{1}{4}$

4. $\frac{1}{128}$

5. $(\sqrt{2})^3$

6. $\sqrt[3]{2}$

7. $\frac{8}{\sqrt{2}}$

Write each of the following as powers of 2.

1. 8

2^3

2. 32

2^5

3. $\frac{1}{4}$

2^{-2}

4. $\frac{1}{128}$

2^{-7}

5. $(\sqrt{2})^3$

$2^{\frac{3}{2}}$

6. $\sqrt[3]{2}$

$2^{\frac{1}{3}}$

7. $\frac{8}{\sqrt{2}}$

$2^{\frac{5}{2}}$

Practice 4

1 Write as a single power of 2:

a $8\sqrt{2}$

b $4\sqrt[3]{2}$

c $\frac{16}{\sqrt{2}}$

d $\frac{\sqrt[3]{2}}{4}$

Problem solving

2 Write as powers of prime numbers:

a $\sqrt[3]{7} \times \sqrt[5]{7}$

b $\sqrt{9} \times \sqrt[4]{9}$

c $\frac{\sqrt[3]{8}}{\sqrt[4]{8}}$

d $\frac{2\sqrt[4]{8}}{\sqrt{2^3}}$

In part **b**, first write 9 as a power of 3.

3 Write as products of powers of prime numbers:

a $\sqrt{6}$

b $\sqrt[3]{14}$

c $\sqrt[4]{162}$

d $\sqrt[4]{100}$

e $\sqrt[4]{50}$

f $\sqrt[6]{72}$

Objective C: Communicating

iii. move between different forms of mathematical representation

In Q4, write radical expressions as exponents to simplify them.

4 Write these expressions as powers of prime numbers. Use the rules of exponents to simplify. Give your answers as powers of prime numbers.

a $\sqrt{2} \times \sqrt[3]{2}$

b $\sqrt[3]{9} \times \sqrt[4]{9}$

c $\sqrt[5]{7} \times \sqrt[10]{7}$

d $\frac{\sqrt{6}}{\sqrt[3]{6}}$

e $\frac{\sqrt[3]{4}}{\sqrt[5]{4}}$

f $\frac{\sqrt[n]{10}}{\sqrt[m]{10}}$

g $\sqrt[n]{a} \times \sqrt[m]{a}$

h $\frac{\sqrt[n]{a}}{\sqrt[m]{a}}$

In parts **g** and **h**, assume a is prime.

Solutions:

Practice 4

$$1 \quad \mathbf{a} \quad 2^{\frac{7}{2}} \quad \mathbf{b} \quad 2^{\frac{7}{3}} \quad \mathbf{c} \quad 2^{\frac{7}{2}} \quad \mathbf{d} \quad 2^{-\frac{5}{3}}$$

$$2 \quad \mathbf{a} \quad 7^{\frac{8}{15}} \quad \mathbf{b} \quad 3^{\frac{3}{2}} \quad \mathbf{c} \quad 2^{\frac{1}{4}} \quad \mathbf{d} \quad 2^{\frac{1}{4}}$$

$$3 \quad \mathbf{a} \quad 2^{\frac{1}{2}} \times 3^{\frac{1}{2}} \quad \mathbf{b} \quad 2^{\frac{1}{3}} \times 7^{\frac{1}{3}} \quad \mathbf{c} \quad 3 \times 2^{\frac{1}{4}} \quad \mathbf{d} \quad 2^{\frac{1}{2}} \times 5^{\frac{1}{2}}$$

$$\mathbf{e} \quad 2^{\frac{1}{4}} \times 5^{\frac{1}{2}} \quad \mathbf{f} \quad 2^{\frac{1}{2}} \times 3^{\frac{1}{3}}$$

$$4 \quad \mathbf{a} \quad 2^{\frac{5}{6}} \quad \mathbf{b} \quad 3^{\frac{7}{6}} \quad \mathbf{c} \quad 7^{\frac{3}{10}} \quad \mathbf{d} \quad 2^{\frac{1}{6}} 3^{\frac{1}{6}}$$

$$\mathbf{e} \quad 2^{\frac{4}{15}} \quad \mathbf{f} \quad 2^{\left(\frac{1}{n} - \frac{1}{m}\right)} \times 5^{\left(\frac{1}{n} - \frac{1}{m}\right)} = 2^{\left(\frac{m-n}{mn}\right)} \times 5^{\left(\frac{m-n}{mn}\right)}$$

$$\mathbf{g} \quad a^{\left(\frac{1}{n} + \frac{1}{m}\right)} = a^{\left(\frac{m+n}{mn}\right)} \quad \mathbf{h} \quad a^{\left(\frac{1}{n} - \frac{1}{m}\right)} = a^{\left(\frac{m-n}{mn}\right)}$$

When radicals have the same radicand but different indices, write them with fractional exponents to simplify them. For example:

- $\sqrt[n]{a} \times \sqrt[m]{a} = a^{\frac{1}{n}} \times a^{\frac{1}{m}} = a^{\frac{1}{n} + \frac{1}{m}} = a^{\frac{n+m}{nm}}$
- $\frac{\sqrt[n]{a}}{\sqrt[m]{a}} = \frac{a^{\frac{1}{n}}}{a^{\frac{1}{m}}} = a^{\frac{1}{n} - \frac{1}{m}} = a^{\frac{n-m}{nm}}$

The radicand is the number under the root sign.

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Example 5

Simplify $\frac{\sqrt[4]{8}}{\sqrt{2}}$

$$\frac{\sqrt[4]{8}}{\sqrt{2}} = \frac{\sqrt[4]{2^3}}{\sqrt{2}}$$

Write the radicands as exponents with the same base.

$$= \frac{2^{\frac{3}{4}}}{2^{\frac{1}{2}}}$$

$$= 2^{\frac{3}{4} - \frac{1}{2}}$$

Give the answer as a radical, with prime number radicand.

$$= 2^{\frac{1}{4}}$$

$$= \sqrt[4]{2}$$

When radicals have different radicands, but one is a power of the other, write them as exponents with the same base to simplify them.

Example 6

Write $10\sqrt{5} \times 3\sqrt[3]{75}$ as a simplified radical, by expressing the radicands as powers of prime numbers.

$$\begin{aligned}10\sqrt{5} \times 3\sqrt[3]{75} &= 10 \times \sqrt{5} \times 3 \times \sqrt[3]{25 \times 3} \\&= 10 \times 3 \times \sqrt{5} \times \sqrt[3]{5^2 \times 3} \\&= 30 \times 5^{\frac{1}{2}} \times 5^{\frac{2}{3}} \times 3^{\frac{1}{3}} \\&= 30 \times 5^1 \times 3^{\frac{1}{3}} \\&= 150\sqrt[3]{3}\end{aligned}$$

Write the radicands as multiples of primes.

Write radicands using exponents.

Practice 5

1 Simplify these completely, leaving your answer as a simplified radical.

a $\sqrt{2} \times \sqrt[3]{2} \times \sqrt[6]{2}$ **b** $\sqrt[3]{3^2} \times \sqrt[4]{3} \times \sqrt[12]{3}$ **c** $\frac{\sqrt{5}}{\sqrt[3]{5}}$ **d** $\frac{\sqrt[3]{11}}{\sqrt{11}}$ **e** $\frac{\sqrt{7^3}}{\sqrt[3]{7}}$
f $\frac{\sqrt[3]{3^4}}{\sqrt{3}}$ **g** $3\sqrt{2} \times 5\sqrt[3]{2}$ **h** $5\sqrt{3^3} \times 2\sqrt[4]{3}$ **i** $\frac{2\sqrt{5}}{\sqrt[3]{5}}$ **j** $\frac{4\sqrt{3^5}}{8\sqrt{3}}$

2 Write as simplified radicals, with prime power radicands.

a $\frac{4}{\sqrt[3]{2}}$ **b** $\sqrt[4]{12} \times \sqrt[3]{3}$ **c** $\sqrt{12} \times \sqrt[3]{18}$ **d** $\frac{\sqrt[3]{16}}{\sqrt{2}}$ **e** $\frac{\sqrt[3]{7}}{\sqrt{343}}$
f $\frac{\sqrt[4]{125}}{\sqrt[3]{5}}$ **g** $\frac{4\sqrt{6}}{\sqrt{3}}$ **h** $\frac{5\sqrt[4]{125}}{\sqrt[4]{5}}$ **i** $\frac{\sqrt[3]{81}}{\sqrt[3]{24}}$ **j** $\sqrt[5]{49} \times \sqrt[5]{343}$

Problem solving

3 Solve these equations, where the unknown is always a power of 3.

a $\frac{\sqrt{27}}{x} = \sqrt{3}$ **b** $\frac{y}{3} = \frac{1}{\sqrt[3]{9}}$ **c** $\frac{\sqrt[3]{9}}{z} = \sqrt[6]{3}$

Solutions:

Practice 5

1 a 2	b 3	c $\sqrt[6]{5}$	d $\frac{1}{\sqrt[6]{11}}$	e 7
f $\sqrt[6]{3^5}$	g $15\sqrt[6]{2^5}$	h $10\sqrt[4]{3^7}$	i $2\sqrt[6]{5}$	j $\frac{9}{2}$
2 a $\sqrt[3]{2^5}$	b $\sqrt[12]{3^7} \times \sqrt{2}$	c $\sqrt[3]{2^4} \times \sqrt[6]{3^7}$	d $\sqrt[6]{2^5}$	e $\frac{1}{\sqrt[6]{7^7}}$
f $\sqrt[12]{5^5}$	g $4\sqrt{2}$	h $5\sqrt{5}$	i $\frac{3}{2}$	j 7
3 a $x = 3$	b $y = \sqrt[3]{3}$	c $z = \sqrt{3}$		

Summary

For any positive real number x , and integer n , where $n \neq 0$, $x^{\frac{1}{n}} = \sqrt[n]{x}$.

For any positive real number x , and integers m and n , where $n \neq 0$, $x^{\frac{m}{n}} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$.

For $a, b \geq 0$: $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$.

For $a \geq 0, b > 0$: $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$.

For $a \geq 0, n > 0$: $\sqrt[n]{a^n} = a$.

The radicand is the number under the root sign.

When radicals have the same radicand but different indices, write them with fractional exponents to simplify them.

For example:

$$\sqrt[n]{a} \times \sqrt[m]{a} = a^{\frac{1}{n}} \times a^{\frac{1}{m}} = a^{\frac{n+m}{nm}}$$

$$\frac{\sqrt[n]{a}}{\sqrt[m]{a}} = \frac{a^{\frac{1}{n}}}{a^{\frac{1}{m}}} = a^{\frac{1}{n} - \frac{1}{m}} = a^{\frac{n-m}{nm}}$$

When radicals have different radicands, but one is a power of the other, write them as exponents with the same base to simplify them.

When radicals have different radicands, but one is *not* a power of the other, then they can not be simplified into one radical.



Optional Further Practice



Evaluate:

a) $64^{\frac{2}{3}}$

b) $81^{-\frac{1}{4}}$

Simplify:

a) $(64a^6)^{\frac{1}{2}}$

b) $\frac{6x^2y^{-2}}{\sqrt[3]{8x^{-3}}}$

Simplify the following (factor or express as a single fraction, or a single term):

a $\left(\frac{4x^2}{5y^4}\right)^2 \cdot (2x^3y)^3$

Level:

1 – 2

b $x^{-1} + y^{-1}$

3 – 4

c $\frac{2^{n-3} \cdot 8^{n+1}}{2^{2n-1} \cdot 4^{2-n}}$

5 – 6

d $\frac{\left(a^{\frac{1}{3}} \cdot b^{\frac{1}{2}}\right)^{-6}}{\sqrt[4]{a^8 \cdot b^9}}$

5 – 6

Solutions:

Evaluate:

$$\begin{aligned} \text{a) } 64^{\frac{2}{3}} &= (4^3)^{\frac{2}{3}} \\ &= 4^2 \\ &= \boxed{16} \end{aligned}$$

$$\begin{aligned} \text{b) } 81^{-\frac{1}{4}} &= \frac{1}{(81)^{\frac{1}{4}}} = \frac{1}{(3^4)^{\frac{1}{4}}} \\ &= \boxed{\frac{1}{3}} \end{aligned}$$

Simplify:

$$\begin{aligned} \text{a) } (64a^6)^{\frac{1}{2}} &= (64)^{\frac{1}{2}} \cdot (a^6)^{\frac{1}{2}} \\ &= \boxed{8a^3} \end{aligned}$$

$$\begin{aligned} \text{b) } \frac{6x^2y^{-2}}{\sqrt[3]{8x^{-3}}} &= \frac{6x^2y^{-2}}{(8x^{-3})^{\frac{1}{3}}} = \frac{6x^2y^{-2}}{2x^{-1}} \\ &= \boxed{\frac{3x^3}{y^2}} \end{aligned}$$

Review: Simplify the following (factor or express as a single fraction, or a single term):

$$\text{a) } \left(\frac{4x^2}{5y^4}\right)^2 \cdot (2x^3y)^3 = \frac{16x^4}{25y^8} \cdot 8x^9y^3 = \boxed{\frac{128x^{13}}{25y^5}}$$

Level:

1-2

$$\text{b) } x^{-1} + y^{-1} = \frac{1}{x} + \frac{1}{y} = \frac{1}{x} \frac{y}{y} + \frac{1}{y} \frac{x}{x} = \boxed{\frac{y+x}{xy}}$$

3-4

$$\text{c) } \frac{2^{n-3} \cdot 8^{n+1}}{2^{2n-1} \cdot 4^{2-n}} = \frac{2^{n-3} \cdot (2^3)^{n+1}}{2^{2n-1} \cdot (2^2)^{2-n}} = \frac{2^{n-3+3n+3}}{2^{2n-1+4-2n}} = \frac{2^{4n}}{2^3} = \boxed{\frac{2^{4n-3}}{8}}$$

5-6

$$\text{d) } \frac{(a^{\frac{1}{3}} \cdot b^{\frac{1}{2}})^{-6}}{\sqrt[4]{a^8 \cdot b^9}} = \frac{a^{-2} \cdot b^{-3}}{(a^8 b^9)^{\frac{1}{4}}} = \frac{a^{-2} b^{-3}}{a^2 b^{\frac{9}{4}}} = a^{-4} b^{-\frac{21}{4}} = \boxed{\frac{1}{a^4 b^{\frac{21}{4}}}}$$

5-6